$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}.$$

From the above equation, we observe that the direction ratios of a line passing through two given points $(x_1, y_1, z_{1)}$ and (x_2, y_2, z_2) are given by $x_2-x_1, y_2-y_1, z_2-z_1$, which are also given by any three numbers proportional to them and in particular $x_1-x_2, y_1-y_2, z_1-z_2$.

Example 6.24

A straight line passes through the point $(1,2,-3)$ and parallel to $4\hat{i}+5\hat{j}-7\hat{k}$. Find (i) vector equation in parametric form (ii) vector equation in non-parametric form (iii) Cartesian equations of the straight line.

Solution

The required line passes through $(1,2,-3)$. So, the position vector of the point is $\hat{i}+2\hat{j}-3\hat{k}$.

Let $\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{b}=4\hat{i}+5\hat{j}-7\hat{k}$. Then, we have

(i) vector equation of the required straight line in parametric form is $\vec{r}=\vec{a}+t\vec{b},\ t\in\mathbb{R}$.

Therefore, $\vec{r}=(\hat{i}+2\hat{j}-3\hat{k})+t(4\hat{i}+5\hat{j}-7\hat{k}),\ t\in\mathbb{R}$.

(ii) vector equation of the required straight line in non-parametric form is $(\vec{r}-\vec{a})\times\vec{b}=\vec{0}$.

Therefore, $(\vec{r}-(\hat{i}+2\hat{j}-3\hat{k}))\times(4\hat{i}+5\hat{j}-7\hat{k})=\vec{0}$.

(iii) Cartesian equations of the required line are $\frac{x-x_1}{b_1}=\frac{y-y_1}{b_2}=\frac{z-z_1}{b_3}$.

Here, $(x_1, y_1, z_1)=(1,2,-3)$ and direction ratios of the required line are proportional to $4,5,-7$. Therefore, Cartesian equations of the straight line are $\frac{x-1}{4}=\frac{y-2}{5}=\frac{z+3}{-7}$.

Example 6.25

The vector equation in parametric form of a line is $\vec{r}=(3\hat{i}-2\hat{j}+6\hat{k})+t(2\hat{i}-\hat{j}+3\hat{k})$. Find (i) the direction cosines of the straight line (ii) vector equation in non-parametric form of the line (iii) Cartesian equations of the line.

Solution

Comparing the given equation with equation of a straight line $\vec{r}=\vec{a}+t\vec{b}$, we have $\vec{a}=3\hat{i}-2\hat{j}+6\hat{k}$ and $\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$. Therefore,

(i) If $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$, then direction ratios of the straight line are b_1, b_2, b_3. Therefore, direction ratios of the given straight line are proportional to $2,-1,3$, and hence the direction cosines of the given straight line are $\frac{2}{\sqrt{14}},\frac{-1}{\sqrt{14}},\frac{3}{\sqrt{14}}$.

(ii) vector equation of the straight line in non-parametric form is given by $(\vec{r}-\vec{a})\times\vec{b}=\vec{0}$.

Therefore, $(\vec{r}-(3\hat{i}-2\hat{j}+6\hat{k}))\times(2\hat{i}-\hat{j}+3\hat{k})=\vec{0}$.

(iii) Here $(x_1, y_1, z_1)=(3,-2,6)$ and the direction ratios are proportional to $2,-1,3$.

Therefore, Cartesian equations of the straight line are $\frac{x-3}{2}=\frac{y+2}{-1}=\frac{z-6}{3}$.

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